



ACTUATION GROUP
SHELBY, NC 28152
CAGE CODE: 71791

CPS 8281 REV A

SHEET NO. 48 OF 70

ACCEPTANCE TEST REPORT
Airbus A350F Motor Control Unit 1 (MCU1)

CWC P/N 190700-1

Serial Number of MCU1: _____

Revision Letter of Assembly Part List: _____

Test Equipment Calibration Check: _____

Pass

Fail

Test Software No.: _____

Rev.

Test Performed By: _____

(Signature)

Test Data Verified By: _____

(Signature)

Start Date: _____

End Date:

Environmental Conditions: _____ Temperature

°C

Humidity

% Relative

REMARKS:



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Motor Control Unit 1 P/N 190700-1

S/N _____

Ground Bonding
Test 5.2

Date: _____

Firmware Version: _____

Ground Bonding Resistance

Pass _____ Fail _____

Test Data

Location	Measured Resistance Under Load (mΩ)	Limit	Pass/Fail
1-2		≤10 mΩ	
1-3		≤10 mΩ	
1-4		≤10 mΩ	
5-2		≤10 mΩ	
5-3		≤10 mΩ	
5-4		≤10 mΩ	
J1 Pin D - 2		≤20 mΩ	
J1 Pin D - 3		≤20 mΩ	
J1 Pin D - 4		≤20 mΩ	



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Motor Control Unit 1 P/N 190700-1

S/N _____

Dielectric Test
Test 5.3

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Leakage Current (≤1 mA)

Pass _____ Fail _____

Visual Inspection (No damage to Location 1)

Pass _____ Fail _____

Test Data – Leakage Current (mA)

	TG1	TG2	TG3	TG4	TG5					
CH_GND	1500VAC	500VDC								
TG1										
TG2	500VDC									
TG3	500VDC	500VDC								
TG4	500VDC	500VDC	500VDC							
TG5	500VDC	500VDC	500VDC	500VDC						
CH_GND										
TG1										
TG2										
TG3										
TG4										
TG5										

Instances of insulation breakdown or failure:



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Motor Control Unit 1 P/N 190700-1 S/N _____

Galvanic Insulation Resistance
Test 5.4

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Baseline Insulation Resistance (≥10 MΩ)

Pass _____ Fail _____

Visual Inspection (no damage to Location 1)

Pass _____ Fail _____

Test Data – Insulation Resistance (MΩ)

	TG1	TG2	TG3	TG4	TG5					
CH_GND	500	45								
TG1		45	45	45	45					
TG2	45		500	500	45					
TG3	45	45		45	45					
TG4	45	45	45		45					
TG5	45	45	45	45						
CH_GND										
TG1										
TG2										
TG3										
TG4										
TG5										

Any observed insulation resistance anomalies or breakdowns:



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Motor Control Unit 1 P/N 190700-1 S/N _____

Test of Presence (Breakdown Voltage) of Lightning Protection Devices
Test 5.5

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Test No.	Pin -	Pin +	V _{BR} Test Limit, Max (V)	I _{Test} (mA)	Recorded V _{BR} (V)	Pass/Fail
1	J2 Pin 51	J2 Pin 54	130	1		
2	J2 Pin 51	J2 Pin 50	130	1		
3	J2 Pin 50	J2 Pin 45	107.5	1		
4	J2 Pin 51	J2 Pin 35	8.8	10		
5	J2 Pin 51	J2 Pin 43	8.8	10		
6	J2 Pin 10	J2 Pin 17	8.8	10		
7	J2 Pin 19	J2 Pin 18	50.8	1		
8	J2 Pin 51	J2 Pin 45	50.8	1		
9	J2 Pin 14	J2 Pin 22	18.4	1		
10	J2 Pin 9	J2 Pin 15	18.4	1		
11	J1 Pin B	J1 Pin A	487	10		
12	J1 Pin C	J1 Pin B	487	10		
13	J1 Pin C	J1 Pin A	487	10		



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Motor Control Unit 1 P/N 190700-1 S/N _____

Pin-to-pin Resistance Test
Test 5.6

Date: _____

Firmware Version: _____

Resistance Limits ($\geq 10\text{ M}\Omega$) **Pass** ____ **Fail** ____

Test Data

Isolation Group 1			Recorded Resistance (Ω)	Pass/Fail
TP+	TP-	Min Resistance (Ω)		
J1 Pin A	J1 Pin B	10M		
	J1 Pin C	10M		
	J3 Pin A	10M		
	J3 Pin B	10M		
	J3 Pin C	10M		
J1 Pin B	J1 Pin C	10M		
	J3 Pin A	10M		
	J3 Pin B	10M		
	J3 Pin C	10M		
J1 Pin C	J3 Pin A	10M		
	J3 Pin B	10M		
	J3 Pin C	10M		
J3 Pin A	J3 Pin B	10M		
	J3 Pin C	10M		
J3 Pin B	J3 Pin C	10M		

Isolation Group 2 (Connector J2)				
TP+	TP-	Min Resistance (Ω)		
J2 Pin 9	J2 Pin 15	10M		
	J2 Pin 16	10M		
	J2 Pin 23	10M		
	J2 Pin 24	10M		
	J2 Pin 31	10M		
	J2 Pin 14	10M		
	J2 Pin 21	10M		



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	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 15	J2 Pin 16	10M		
	J2 Pin 23	10M		
	J2 Pin 24	10M		
	J2 Pin 31	10M		
	J2 Pin 14	10M		
	J2 Pin 21	10M		
	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 16	J2 Pin 23	10M		
	J2 Pin 24	10M		
	J2 Pin 31	10M		
	J2 Pin 14	10M		
	J2 Pin 21	10M		
	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 23	J2 Pin 24	10M		
	J2 Pin 31	10M		
	J2 Pin 14	10M		
	J2 Pin 21	10M		
	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 24	J2 Pin 31	10M		
	J2 Pin 14	10M		
	J2 Pin 21	10M		
	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 31	J2 Pin 14	10M		
	J2 Pin 21	10M		
	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 14	J2 Pin 21	10M		
	J2 Pin 30	10M		



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	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 21	J2 Pin 30	10M		
	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 30	J2 Pin 22	10M		
	J2 Pin 29	10M		
J2 Pin 22	J2 Pin 29	10M		

Isolation Group 3 (Connector J2)				
TP+	TP-	Min Resistance (Ω)		
J2 Pin 40	J2 Pin 41	10M		
	J2 Pin 32	10M		
	J2 Pin 33	10M		
	J2 Pin 47	10M		
	J2 Pin 53	10M		
	J2 Pin 48	10M		
J2 Pin 41	J2 Pin 32	10M		
	J2 Pin 33	10M		
	J2 Pin 47	10M		
	J2 Pin 53	10M		
	J2 Pin 48	10M		
J2 Pin 32	J2 Pin 33	10M		
	J2 Pin 47	10M		
	J2 Pin 53	10M		
	J2 Pin 48	10M		
J2 Pin 33	J2 Pin 47	10M		
	J2 Pin 53	10M		
	J2 Pin 48	10M		
J2 Pin 47	J2 Pin 53	10M		
	J2 Pin 48	10M		
J2 Pin 53	J2 Pin 48	10M		

Isolation Group 4 (Connector J2)				
TP+	TP-	Min Resistance (Ω)		
J2 Pin 35	J2 Pin 43	10M		
	J2 Pin 55	10M		



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	J2 Pin 50	10M		
	J2 Pin 46	10M		
	J2 Pin 45	10M		
	J2 Pin 52	10M		
J2 Pin 43	J2 Pin 55	10M		
	J2 Pin 50	10M		
	J2 Pin 46	10M		
	J2 Pin 45	10M		
	J2 Pin 52	10M		
J2 Pin 55	J2 Pin 50	10M		
	J2 Pin 46	10M		
	J2 Pin 45	10M		
	J2 Pin 52	10M		
J2 Pin 50	J2 Pin 46	10M		
	J2 Pin 45	10M		
	J2 Pin 52	10M		
J2 Pin 46	J2 Pin 45	10M		
	J2 Pin 52	10M		
J2 Pin 45	J2 Pin 52	10M		

Isolation Group 5 (Connector J2)				
TP+	TP-	Min Resistance (Ω)		
J2 Pin 18	J2 Pin 19	10M		
	J2 Pin 10	10M		
	J2 Pin 17	10M		
J2 Pin 19	J2 Pin 10	10M		
	J2 Pin 17	10M		
J2 Pin 10	J2 Pin 17	10M		



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Inrush Current Test
Test 5.7

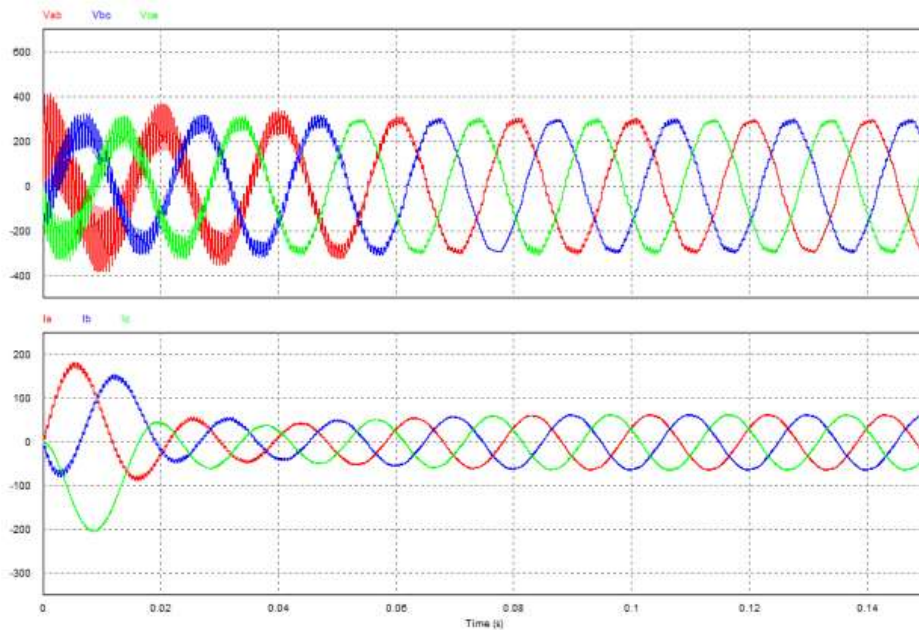
Date: _____

Inrush current envelope

Pass _____ Fail _____

Test Data

Oscilloscope screenshots



Note: this is an example of the data to be displayed / recorded (not the pass/fail envelope).



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Standby Current Draw Test
Test 5.8

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Test Data

Measurement	Recorded Value	Pass/Fail Limit	Pass/Fail
Phase A current		<1A	
Phase B current		<1A	
Phase C current		<1A	
Health Status		READY	



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Motor Control Unit 1 P/N 190700-1

S/N _____

Firmware Version Test
Test 5.9

Date: _____

Firmware Version: _____

Firmware Result

Pass ____ Fail _____

Test Data

Expected Firmware Version	Retrieved Firmware Version	Expected Firmware Checksum	Retrieved Firmware Checksum

Mismatch/anomalies



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Motor Control Unit 1 P/N 190700-1 S/N _____

**Input/Output (I/O) Range Test – STO Input/Output
Test 5.10**

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Measurement	Expected	Actual	Pass/Fail
$V_{threshold\ up}$	$\geq 9\ V$		
$V_{threshold\ down}$	$\leq 5.5\ V$		
STO Output Impedance (when sufficient input voltage is present)	$< 75\Omega$		
STO Output Impedance (no input voltage)	$\geq 1M\Omega$		



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Motor Control Unit 1 P/N 190700-1 S/N _____

Input/Output (I/O) Range Test – MCU2 Enable / MCU2 Status Discretes
Test 5.11

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Measurement	Recorded Value	Pass/Fail Limit	Pass/Fail
V _{threshold_up}		$\geq 0.9 \text{ V}$	
V _{threshold_down}		$\leq 1.2 \text{ V}$	
Voltage on J2 Pin 47 (Health Status Ready)		$4.4 \text{ V} \leq \text{STS} \leq 5 \text{ V}$	
Voltage on J2 Pin 47 (Health Status Not Ready)		$< 0.2 \text{ V}$	



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Input/Output (I/O) Range Test – ARINC 429
Test 5.12

Date: _____

Firmware Version: _____

OVERALL

Pass_____Fail

_____Test Data

Measurement	Recorded Value	Pass/Fail Limit	Pass/Fail
A429_DSCS_TX High Voltage		Voltage ≥ 9 V	
A429_DSCS_TX Low Voltage		Voltage ≤ -9 V	
A429_DSCS_TX Slew Rate		4-8 V μs^{-1}	
A429_MCU2_TX High Voltage		Voltage ≥ 9 V	
A429_MCU2_TX Low Voltage		Voltage ≤ -9 V	
A429_MCU2_TX Slew Rate		4-8 V μs^{-1}	



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Motor Control Unit 1 P/N 190700-1 S/N _____

Input/Output Range Test – Resolver
Test 5.13

Time: _____

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Test Data – Voltage and Frequencies

Measurement	Pass/Fail Limits	Actual (°)	Pass/Fail
GRA Resolver excitation frequency	4.65–4.83 kHz		
GRA Resolver excitation voltage	4.5–5.5 Vrms		
EPDU Resolver excitation frequency	19.00–21.00 kHz		
EPDU Resolver excitation voltage	3.6–4.4 Vrms		

Test Data – Resolver Reading Accuracy (using Emulator)

Measurement	Reported GRA Angle Pass Fail Limits	Actual (°)	Pass/Fail
GRA reported angle for 0° emulated angle	±0.5°		
GRA reported angle for 216.3165° (-143.6835°) emulated angle	50 ±0.5°		
GRA reported angle for 72.6329° (-287.3671°) emulated angle	100 ±0.5°		

Test Data – Resolver Reading Accuracy (Alternative Method using Real GRAPR)

Measurement	Actual GRAPR Angle (°)	[1] Actual GRAPR Angle ÷ 2.873671	Reported GRA Angle (°)	Pass/Fail Limit (±0.5° of [1])	Pass/Fail
GRAPR Angle between - 0° and -10°					
GRAPR Angle between - 140° and -150°					
GRAPR Angle between - 277° and -287°					



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Motor Control Unit 1 P/N 190700-1 S/N _____

**Input/Output Range Test – EPDU Temperature
Test 5.14**

Date: _____

Firmware Version: _____

MCU1 Temperature Readings

Pass_____Fail_____

Test Data MCU1 Temperature Readings

Temperature Setting (°C)	Temperature Reading (°C)	Pass/Fail Limit (°C)	Pass / Fail
+25		+15 to +35	
-40		-50 to -30	
+110		+100 to +120	



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Motor Control Unit 1 P/N 190700-1 S/N _____

Safe Torque Off (STO) Test
Test 5.15

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Test Data

Status	Open Command Issued	EPDU STO Function Status	Phase A Input Current (A)	Phase A Input Current Pass/Fail Limit	Pass / Fail
STO Active #1	☐	☐ Active		<0.5A	
STO disengaged	☐	☐ Inactive		>1A	
STO Active #2	☐	☐ Active		<0.5A	



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Motor Control Unit 1 P/N 190700-1

S/N _____

Efficiency Test
Test 5.16

Time: _____

Date: _____

Firmware Version: _____

OVERALL

Pass ____ Fail ____

Test Data

Output Load (N-m)	Output Load (kW)	Input Load (kW)	Efficiency (%)	Pass / Fail Criteria	Pass / Fail
1 N-m				≥80%	
2.5 N-m				≥80%	



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Motor Control Unit 1 P/N 190700-1

S/N _____

Functional Test
Test 5.17

Time: _____

Date: _____

Firmware Version: _____

OVERALL

Pass _____ Fail _____

Test Data – Loaded Open Cycle

Measured Parameter	Measured	Pass/Fail Limit	Pass/Fail
Max Phase A current		<12.5Arms	
Max Phase B current		<12.5Arms	
Max Phase C current		<12.5Arms	
Motor Speed at Max Torque		xx rpm $\pm$ xx rpm ($\pm 10\%$)	
Full cycle complete?		Complete	

Test Data – Loaded Close Cycle

Measured Parameter	Measured	Pass/Fail Limit	Pass/Fail
Max Phase A current		<4Arms	
Max Phase B current		<4Arms	
Max Phase C current		<4Arms	
Motor Speed at Max Torque		xx rpm $\pm$ xx rpm ($\pm 10\%$)	
Full cycle complete?		Complete	



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Test Data – Jam (Open Direction)

	Measured Value	Pass/Fail Limit	Pass/Fail
Current		$24 \pm 2.4 \text{ A}$	
Duration		$2 \pm 0.5 \text{ s}$	

Test Data – Jam (Close Direction)

	Measured Value	Pass/Fail Limit	Pass/Fail
Current		$7 \pm 1.05 \text{ A}$	
Duration		$2 \pm 0.5 \text{ s}$	



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Motor Control Unit 1 P/N 190700-1

S/N _____

Date: _____

Inspector: _____

190700-ICD Rev _____

Marking and Nameplate

Test 5.18

Document	Revision Level

OVERALL

Pass _____ Fail _____

Correct Information (All markings correct) Pass _____ Fail _____

Proper Placement (Locations correct) Pass _____ Fail _____

Nameplate Legibility (Clearly legible) Pass _____ Fail _____

Nameplate Security (No peeling, lifting, or Misalignment.) Pass _____ Fail _____

Interface Dimensions

Test 5.19

Compliance with ICD Drawings (Interface dimensions within tolerance) Pass _____ Fail _____

Weight Check

Test 5.20

Target Weight (≤ 10.7 kg) Pass _____ Fail _____

Weight: _____ kg



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Finish and Workmanship
Test 5.21

Visual Inspection

(No visible flaws)

Pass _____ Fail _____

Pin check

(No broken or bent pins)

Pass _____ Fail _____

Notes: